Complexity Report - Transportation Problem

1. Introduction

The goal of this project is to understand how the choice of the initial solution method affects the total computation time when solving a transportation problem.

In practice, solving the problem is done in two steps: we first build a feasible solution, then we improve it using an optimization method. This leads to two complete pipelines that we want to compare:

- Northwest + Stepping Stone
- Balas-Hammer + Stepping Stone

2. Data handling

Before comparing the algorithms, we first need to generate valid transportation problems. For a given size n, we build:

- a cost matrix of size n × n
- a supply vector of size n
- a demand vector of size n

To ensure the problem is feasible, the total supply is equal to the total demand. We created random problems to ensure that we get a good idea of the average complexity we would get for any problems in real situations, as they are rarely ideal

3. Initial solutions

At this stage, we build a first feasible solution. The goal is not to be optimal yet, but simply to satisfy all constraints.

North-West

The North-West method builds a solution by filling the matrix starting from the top-left corner. At each step, it assigns as much flow as possible, then moves either to the right or downward. Because each row and each column is visited at most once, the algorithm progresses through the matrix in a single pass. This means the number of operations grows with the number of cells. And we end up with a complexity of $O(n^2)$. So, this method is extremely fast.

However, it completely ignores the costs. As a result, the solution obtained is often far from optimal. This is important, because it directly impacts the next phase: a bad starting point means more work during optimization.

Balas-Hammer

The Balas-Hammer method takes a different approach: it tries to make better decisions from the start. At each step, it computes penalties for each row and column based on costs, then selects the most promising allocation. Unlike North-West, it repeatedly scans parts of the matrix to recompute these penalties. So instead of a simple pass, we have multiple scans of size n^2, repeated about n times. This leads to a higher cost, $O(n^3)$. So, Balas-Hammer is clearly slower at this stage. However, the solution obtained is already closer to the optimal solution, which will reduce the work needed later.

4. Stepping Stone with potential (optimization)

Once an initial solution is built, we improve it using the Stepping Stone method. The goal is to reduce the total cost by modifying flows along cycles. At each iteration:

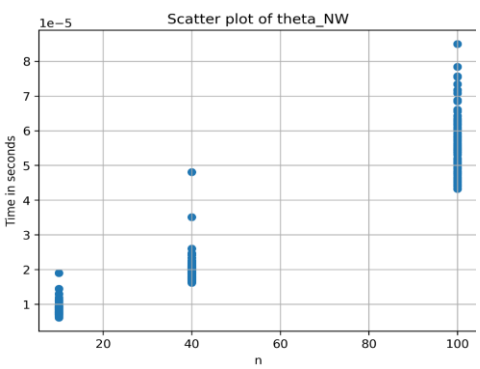
- we test candidate cells that are not currently used
- we build a cycle
- we compute the gain
- we update the solution

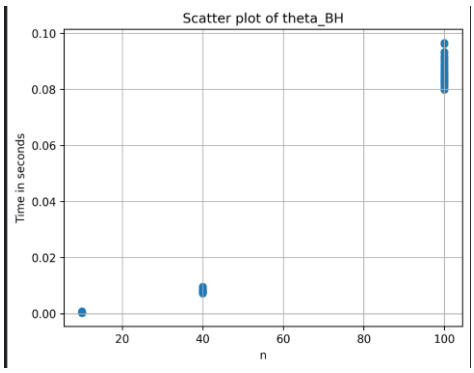
This process is repeated until no improvement is possible. Two things increase with n, the number of possible cycles to test and the number of iterations required. So, the cost grows very quickly in practice. This phase is the most expensive one, and it dominates the total runtime, especially when the starting solution is poor.

5. Experimental study and scatter plots

To observe this behavior, we generate 100 random problems for each value of n. For each run, we measure:

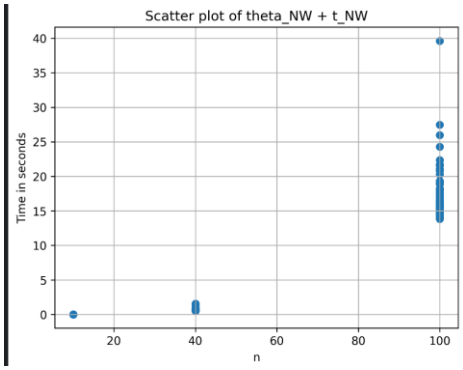
- θNW(n): time to build the North-West solution
- θBH(n): time to build the Balas-Hammer solution
- tNW(n).: optimization time starting from North-West
- TBH(n): optimization time starting from Balas-Hammer

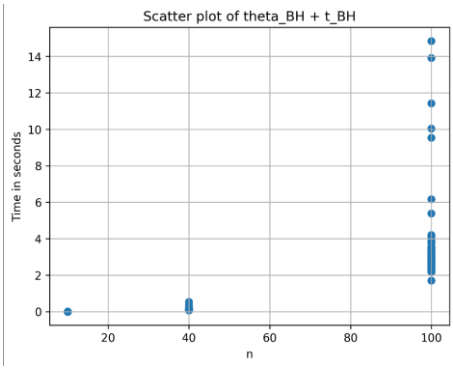




We observe that θNW(n) remains extremely small and almost constant. This confirms that North-West is both fast and very stable. For θBH(n), the values are larger and slightly more dispersed. This reflects the additional computations required by Balas-Hammer. At this stage,

North-West is clearly faster. But as we discussed earlier, this isn't the entire story. We will do the same test, but, adding the stepping stone algorithm execution.

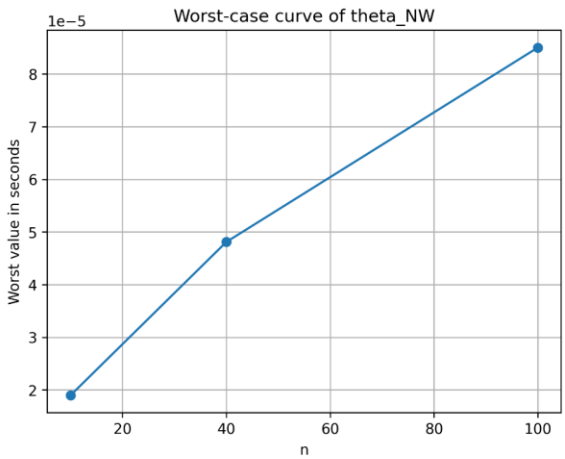


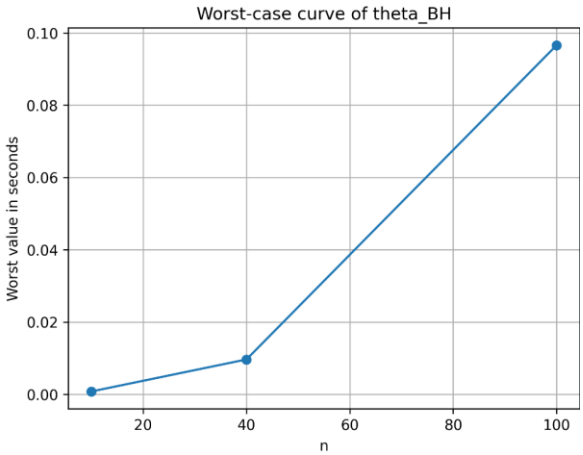


We observe that the Northwest pipeline shows very high dispersion, especially for larger values of n. For n = 100, most runs are already around 15 to 25 seconds, but some outliers reach nearly 40 seconds. This indicates that the optimization phase can become extremely expensive depending on the instance. Meanwhile for Balas Hammer, the cluster is around 2 to 4 seconds, with outliers up to 15 seconds. This is much faster than Northwest.

6. Worst case curves

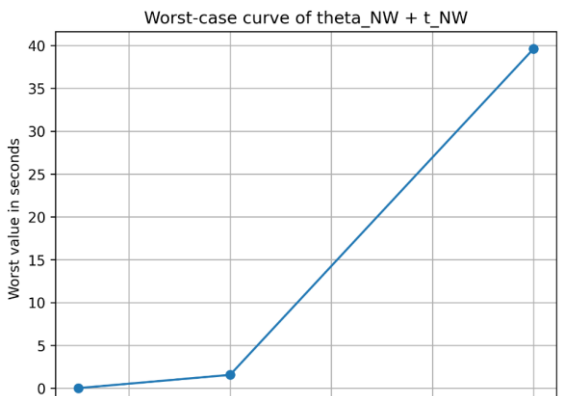
To better understand extreme behaviors, we will focus on the worst case only. Meaning the maximum time, we've observed among the 100:

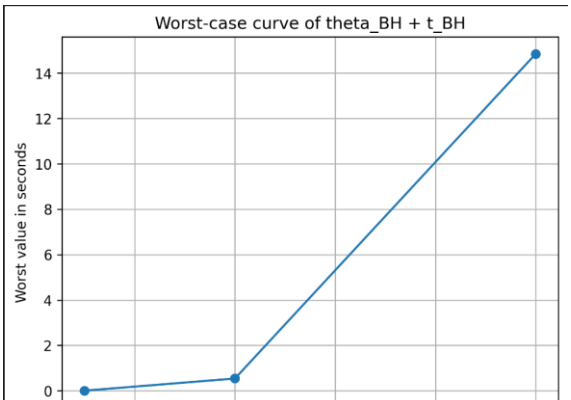




For θNW(n), the curve remains flat and very small. This confirms that the construction time of the Northwest method is negligible, even in the worst case, and is consistent with linear complexity. For θBH(n), the growth is slightly higher and steeper due to additional computations. However, the cost remains negligible compared to the Stepping Stone algorithm. This behavior is consistent with a quadratic complexity.

Now, we will consider the worst-case total time for both methods, with Stepping stone:

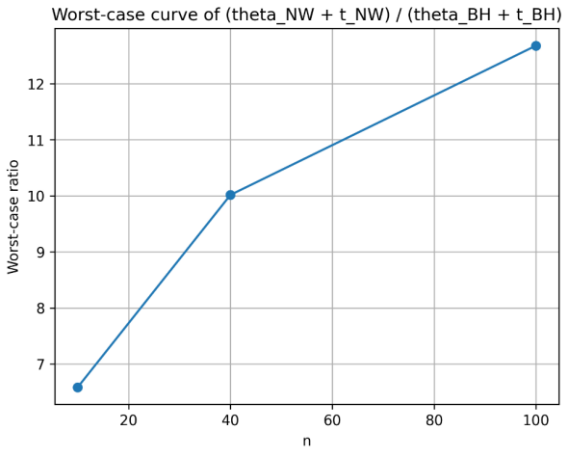




This time for θNW(n) the curve increases very sharply. Even at n = 40, the worst-case time is higher than for the Balas-Hammer method, and for n = 100, it reaches 40 seconds. This behavior suggests a polynomial complexity with a relatively high degree (k > 2), dominated by the optimization phase. For the Balas-Hammer pipeline, the increase is also visible but remains much lower. At n = 40, the worst-case is around 0.5 seconds, and at n = 100, it is approximately 15 seconds.

This confirms that Balas-Hammer keeps the total computation time more controlled, even in extreme cases. Its behavior is also consistent with a polynomial complexity, but with a significantly smaller growth rate.

7. Worst case comparison



Here, we see that the ratio never goes below 1 and even stays well above it, meaning that it is much more expensive to use the Northwest method instead of Balas-Hammer before optimization. As n increases, the curve continues to grow, although more slowly. This indicates that the performance gap appears early and remains significant as the problem size increases. This means that Balas-Hammer consistently provides a better starting solution, leading to a lower overall computation time.

8. Conclusion

This study shows that comparing only the construction phase is not sufficient to evaluate the performance of the two methods. The North-West method is extremely fast to compute and scales very well at this stage. However, it often produces less efficient initial solutions, which increases the cost of the optimization phase. The Balas-Hammer method is slower during construction, but it generates better initial solutions. This reduces the work required during optimization and leads to a lower total computation time in most cases. Even for moderate problem sizes (up to n = 100), the results clearly show that the optimization phase dominates the total runtime. The quality of the initial solution therefore plays a key role in overall performance. As the problem size increases, the difference between the two approaches becomes more visible, making Balas-Hammer a more efficient choice for solving transportation problems in practice.